A Review on

Real Time Data Stream Classification and Adapting To Various Concept Drift Scenarios

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Abstract*—Data streams are viewed as a sequence of relational tuples (e.g., sensor readings,call records, web page visits) that continuously arrive at time-varying and possibly unbound streams.These data streams are potentially huge in size and thus it is impossible to process many data mining techniques and approaches. Classification techniques fail to successfully process data streams because of two factors: their overwhelming volume and their distinctive feature known as concept drift. Concept drift is a term used to describe changes in the learned structure that occur over time. The occurance of concept drift leads to a drastic drop in classification accuracy. The recognition of concept drift in data streams has led to sliding-window approaches also different approaches to mining data streams with concept drift include instance selection methods, drift detection, ensemble classifiers, option trees and using Hoeffding boundaries to estimate classifier performance. This paper describes the various types of concept drifts that affect the data examples and discusses various approaches inorder to handle concept drift scenarios.The aim of this paper is to review and compare single classifier and ensemble approaches to data stream mining respectively.

Index Terms : Data stream mining; concept drift; ensemble classifier; dynamic environment.

I. INTRODUCTION

Most of the previous and current research in data stream mining is carried out in static environments, wherein a complete dataset is presented to the learning algorithm. These data were usually electronically stored and, if needed, was easily accessed by the learning algorithms several times. However, the target concepts which should be learned were always fixed. Over these years, many solutions to the static classification have been developed and several quite accurate classifiers are now available in large scale. However, in some of the newest and latest applications, learning algorithms work in dynamic environments, where a huge amount of data are continuously generated.Traffic management Sensor networks, monitoring, web log analysis or telecommunication are examples of such applications. In these dynamic environments, incoming data form a data stream are characterized by huge number of instances and they arrive rapidly, which often requires real-time and quick response. Compared to static environments, the processing of data streams implies a variety of new requirements for learning algorithms, such as constraints on memory usage, restricted learning, and one scan of incoming instances and testing time.

Concept drift occurs when the concept about which data are being collected shifts from time to time after a minimum stability period. Such changes are reflected in incoming instances and decreases the accuracy of classifiers learned from past training tuples(input data examples). Examples of real life concept drifts include monitoring systems, financial fraud detection spam categorization weather predictions and customer preferences.

Changes of target concepts are categorized into sudden, incremental, gradual, recurring, blip or noise drifts. Figure 1. Shows the six basic types of drifts. The first plot (Sudden) shows abrupt changes that instantly and irreversibly change the data examples of respective class. The next two plots (Incremental and Gradual) illustrate changes that take place slowly over time. Incremental drift occurs when data example slowly change their values over time, and gradual drift occurs when the change in data example involves the class distribution of various data.

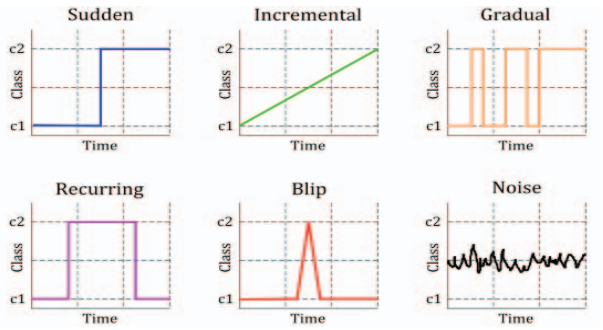


Fig. 1. Different types of Concept Drifts

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Some researchers do not differentiate these two types of drift and use the terms incremental gradual and as synonyms. The left-bottom plot (Recurring) represents changes in data example that are only temporary and are reverted after some time. Some researchers refer such changes as *local drift*. Local drift happens when several data are expected to switch over time and reappear at irregular time intervals.This drift is not certainly periodic in nature. The fifth plot (Blip) represents a "rare event", which can be considered as an outlier in a static distribution. The last plot in Figure 1. represents random changes in data examples, which should be filtered out efficiently. A good classifier must learn incrementally and adapt to such changes.

In recent years, several approaches that try to handle various concept drift scenarios have been proposed effictively, including mainly sliding window approaches [1],adaptive sliding window techniques [2], new online algorithms, special detection techniques,Drift detection methods [4], Hoeffding trees [6], Ensemble drift detection methods and adaptive ensembles [8]. Ensembles are popular approaches for improving classification accuracy in static learning environment, however, they must be generalized for changing environments. Such a generalization can either modify the structure of the ensemble (weaker components are replaced by base classifiers trained on the most recent data examples), updating the aggregation technique (e.g., updating weights in the voting formula).

II. RELATED WORK

A. WINDOWING TECHNIQUE

Windows provide a way that limit the amount of data examples introduced to the learner, thus eliminating those data examples that come from an old concept.

Structure:

The basic windowing algorithm is straightforward in nature. Each data example updates the window and later the classifier is updated by that window itself. In the way it models the forgetting process. In the simplest approach sliding windows are of fixed size and include only the most recent examples from the data stream under consideration. With each new data point the oldest data example that does not fit in the window is thrown away.

Disadvantage:

When fixed size window is used, the user is caught in a tradeoff. In case of small window size the classifier will react quickly to changes, but may loose on accuracy in periods of stability and on the other hand, choosing a large size window will result in increasing accuracy in periods of stability, but will fail to adapt to rapidly changing concept drifts. That is why dynamic ways of modelling the forgetting process have been proposed which is known as Weighted Windows. A simple way of making the forgetting process more dynamic is providing the window with a decay function that assigns a weight to each data example. Older examples receive smaller weights and are treated as less important by the base classifier algorithm. Cohen and Strauss analyzed the use of different decay functions for calculating data stream aggregates.

B. FISH [2]

Zliobaite proposed a family of algorithms which is known as FISH(FISH1,FISH2,FISH3), that uses time and space similarities between data examples as a way of dynamically creating a window. To explain the approach, let us consider an illustrative example, which we present in Figure 2. A binary classification problem is represented by black and white dots as shown below.

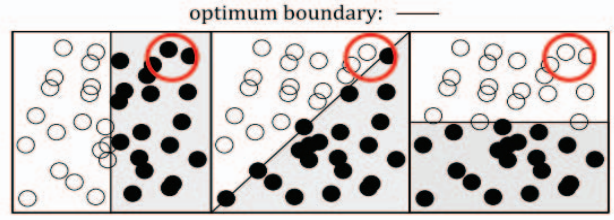


Fig. 2. Rotating hyperplane example (left) initial source S1, (center) source S2 after 45□ rotation, (right) source S3 after 90□ rotation. Black and white dots represent the two classes.

The data generating sources change along with time, gradually rotating the optimal classification hyperplane. For a given fixed space area, shown with a big round circle, the correct class changes as the optimal boundary rotates. The examples shows that similarity in an evolving environment depends on both time and space respectively.

Structure:

For each new example xi in the data stream, FISH3 evaluates the different time/space proportions and window sizes.FISH3 allows to dynamically establish the size of the training window and also provide the proportion between time and space weights. The algorithm's previous version FISH2 takes the time/space proportion as a parameter, while the first algorithm from the family, FISH1, uses a fixed window of the nearest instances and this is the major difference between the two algorithms.

Disadvantage:

FISH3 is an algorithm that always need to iterate through many window sizes and time/space proportions each time performing leave-one-out cross validation. This is a costly process and may be unfeasible for rapid data streams scenarios.

C. ADWIN [3]

Bifet proposed an adapting sliding window algorithm called ADWIN suitable for data streams with sudden drift scenarios. The algorithm keeps a sliding window W with the most recently read data examples. The main idea of ADWIN is as follows: whenever two "large enough" subwindows of W exhibit "distinct enough" averages, one can conclude that the corresponding expected values are not similar, and thus the older portion of the window is dropped.

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Structure:
The ideal part of the algorithm lies in the definition of cut and the test it is used for. The authors stated, the different statistical tests can be used for this purpose, but propose one specific implementation only.

Disvantage:
The verification of all subwindows is very costly process due to the number of possible split points. ADWIN works only for 1-dimensional data only, e.g., the running error. For n-dimensional raw data, a separate window must be maintained for each dimension which results in handling more than one window. This modified model, although costly, reflects the fact that the importance of each feature may change at different pace.

III. SURVEY ON DRIFT DETECTOR

After windowing techniques, another group of algorithms allowing to adapt almost any learner to evolving data streams are drift detectors. Their task is to detect concept drift and alarm the base learner that its model should be rebuilt or updated.

A. DDM [4]

Gama proposed the Drift Detection Method that implies, in each iteration an online classifier predicts the decision class of an example. That prediction can be either true or false, thus for a set of example the error is a random variable. That is why the number of classification errors with a Binomial distribution calculated as given by Equation

$$s_i = \sqrt{p_i(1-p_i)/i} \quad (1)$$

These values are used to calculate and a warning level condition presented in Equation 2. and an alarm level condition presented in Equation 3. Each time a warning level is reached, the data examples are remembered in a separate window.

$$pi + si \geq p_{min} + \alpha \cdot s_{min} \quad (2)$$
$$pi + si \geq p_{min} + \beta \cdot s_{min} \quad (3)$$

The values α and β in the above conditions decide about the confidence levels at which the warning and alarm signals are triggered only then. The authors proposed $\alpha = 2$ and $\beta = 3$, giving approximately 95% confidence of warning and 99% confidence of concept drift.

Disvantage:
DDM works best on data streams with sudden concept drift as gradually changing concept drifts can pass without triggering the alarm level. When no changes are detected, DDM works like a lossless learner constantly enlarging the window size that can lead to the memory limit being exceeded.

B. EDDM [4]

Baena-Garcia et al. proposed a modification of Drift Detection Method called EDDM. The authors use the same warning-alarm mechanism that was proposed by Gama, but instead of using the classifier's error rate, the distance error rate was proposed.

Disvantage:
EDDM works better than DDM for slow gradual drift, but is more sensitive to noise issues. Another drawback of this EDDM is that it considers the threshold values and searches for concept drift when a minimum of 30 errors have occurred. This is necessary to approximate the Binomial distribution by a Normal distribution, but can take a large amount of examples to happen over.

C. HOEFFDING TREES [5]

Decision trees were the first learners to be adapted to data stream mining by using the Hoeffding trees bound. The Hoeffding bound states : with the probability $1 - \delta$, the true mean of a random variable of range R will not differ from the estimated mean after n independent observations by more than:

$$E = \sqrt{R^2 \ln(1/\delta) / 2n} \quad (4)$$

Using this bound, Domingos and Hulten proposed a classifier called Very Fast Decision Tree.Their performance has been compared with traditional decision trees, Naive Bayes, kNN, and ensemble methods respectively. They proved to be much faster and less memory consuming while handling extremely large datasets.

IV. SURVEY ON ENSEMBLE STRATEGIES FOR CHANGING ENVIRONMENTS

Ensemble Classifier are a common way of boosting classification accuracy. Due to their modularity, they provide a natural way of adapting to change by modifying ensemble members also ensemble classifiers mine evolving data streams.
Ensemble algorithms are sets of single classifiers components whose decisions are aggregated by a voting rule. The combined decision of many single classifiers is more accurate than that given by a single component. Studies show that to obtain this accuracy boosting, it is necessary to diversify ensemble members from one other.

A. STREAMING ENSEMBLE ALGORITHM (SEA) [6]

Street and Kim proposed an ensemble method called Streaming Ensemble Algorithm (SEA) that changes its structure to react to changes. They propose a heuristic replacement strategy of the "weakest" expert based on two factors: accuracy and diversity. SEA performed almost as well as a pruned tree on static data sets and much better

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on data sets with concept drift.It showed that SEA performed best when no more than 25 components were used, base classifiers were unpruned, and simple majority voting was used to combine member decisions.

B. ACCURACY WEIGHTED ENSEMBLE(AWE) [7]

In Accuracy Weighted Ensemble (AWE), they train a new classifier C_0 on each incoming data chunk and use that chunk to evaluate all the existing ensemble members to select the best component classifiers.

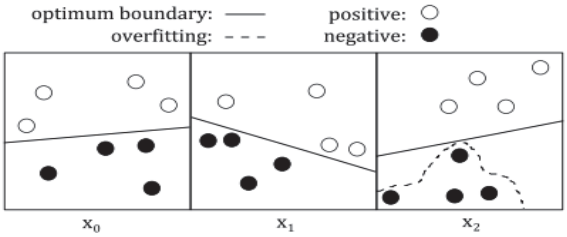


Fig. 3. Data Distribution

The AWE algorithm works well on data streams with reoccurring concepts as well as different types of drift. As with SEA it is crucial to properly define the data chunk size as it determines the ensembles flexibility. AWE will improve its performance gradually over time and is best suited for large data streams.

C. HOEFFDING OPTION TREES AND ASHT BAGGING [8]

Recently, two interesting ensemble methods which use Hoeffding trees were been proposed. that allows each training example to update a set of option nodes rather than just a single leaf. Hoeffding Option Trees (HOT) provide a compact structure that works like a set of weighted classifiers, and just like regular Hoeffding Trees, they are built in an incremental fashion.When compared with Hoeffding Option Trees, ASHT Bagging proves to be more accurate on most data sets, but is extremely time and memory expensive.

D. ACCURACY DIVERSIFIED ENSEMBLE [9]

Accuracy Weighted Ensemble is based on its weighting mechanism, whereas Accuracy Diversified Ensemble (ADE), which not only selects but also updates components according to the current distribution. AWE was designed to use traditional batch classifiers as base learners. Because of this, they have to create ensemble members from single chunks and later only adjust component weights according to the current distribution. This makes the data chunk size a crucial parameter for AWE's performance. Another drawback of AWE is its weight function. Because the algorithm is designed to perform well on cost-sensitive data, the mean square error threshold cuts-off "risky" classifiers. In rapidly changing environments (like the Electricity dataset) this threshold can"mute" all ensemble members causing no class to be predicted. Compared to existing ensemble methods the Accuracy Diversified Ensemble provides a new learning strategy. ADE differs from AWE in weight definition, the use of online base classifiers,bagging and updating components with incoming examples.

E. ACCURACY UPDATED ENSEMBLE (AUE) [9]

The Accuracy Updated Ensemble maintains a weighted pool of component classifiers and predicts the class of incoming examples by aggregating the predictions of components using a weighted voting rule. After each data chunk of examples,a new classifier is created, which substitutes the weakest performing ensemble member.

The performance of each component classifier is evaluated by estimating its expected prediction error on the examples from the most recent data chunk. After substituting the poorest performing component, the remaining ensemble members are updated, i.e., incrementally trained, and their weights are adjusted according to their accuracy. It uses Hoeffding Trees as component classifiers, but the presented algorithm can be considered as a general method, and in principle, one could use other online learning algorithms as base learners.

V. CONCLUSION

In this paper the problem of mining time evolving data streams are discussed. It defined the main characteristics of data streams and discussed different types of changes that occur in streaming data. During the discussion it focused on non-random class definition changes called concept drift. It also reviewed existing single classifier and ensemble approaches to mining data streams with concept drift. The analysis led to the development of a new algorithm called Accuracy Diversified Ensemble, which is based on our critique of the earlier developed Accuracy Weighted Ensemble.

It found that bagging provided time and memory overhead without improving accuracy.Findings concerning the windowed classifier.The mining of data streams with concept drift is growing into a new branch of knowledge discovery, with its own unique research problems. The need for online processing with time and memory constraints forces researchers to focus on resource usage while designing accurate classifiers.

As future work, the plan to carry out additional experiments to analyze the relationship between data chunk size and the performance of bagging in the Accuracy Diversified Ensemble can be done. Further- more, since bagging did not prove to provide additional accuracy to ensemble method, the plan to explore different ways of diversifying ensemble members like the use of heterogeneous base learners or boosting. Finally, in future experiments comparing a larger number of algorithms to take a broader look at the performance of the most recent stream mining techniques can be carried out.

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ACKNOWLEDGMENT

The author would like to thank "Dariusz Brzezinski" for making available all the latest technique applied to mining data stream with concept drift and experimental evaluation and comparison of different approaches towards drift detection and classification.

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